

Lakes Inspection

Lakes in Lake district are connected by canals. The filters, which help to keep water in the lakes clean, were built decades ago, many of them are poorly maintained, and there is a concern that significant contamination may occur in some lakes soon. The management of Lake district has assembled a committee which is going to inspect the lakes.

The committee has an inspection barge at their disposal, and they plan to sail through the lakes and canals to assess and report the state of the lakes on the list.

Traveling through canals in Lake district is limited by the fact that all canals are one-way waterways. Each canal connects two different lakes, each lake can be a start point of none, one, or more than one canal and also it can be an end point of none, one, or more than one canal.

The barge is located in its dock in one of the lakes. The committee will start their inspection journey either in the lake with the barge dock, let us call it B , or in another lake accessible via canals from B , let us call that lake S . From S , the barge will travel through various canals and lakes and it will end its journey again in S . If lake S is going to be other lake than B , it should be chosen in such way that also B is accessible from S , so that the barge can be moved to its dock after the inspection is finished.

It is possible that under these limited conditions the committee will not be able to inspect all affected lakes and that some of the lakes will remain uninspected.

Still, the committee wants to maximize the number of lakes on the list which can be reached during the journey.

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The management decided to help the committee by offering the following **optional** action: They are going to temporarily reverse the direction of one canal, let us call it C , which the committee will point out. The direction of C will be reversed before the barge starts its journey. Immediately after the barge passes through it, this canal C will be closed because of the necessary maintenance of C after the reversal. To allow enough time for the maintenance, C must be the first canal in the barge journey and therefore S must be the lake at the end of C in its current orientation.

If the committee decides to use a canal reversion, it has to choose C , and consequently S , so that the number of lakes which the barge can visit on its journey is maximized. The barge can travel relatively fast, therefore it does not matter if it sails through some lakes or canals more than once.

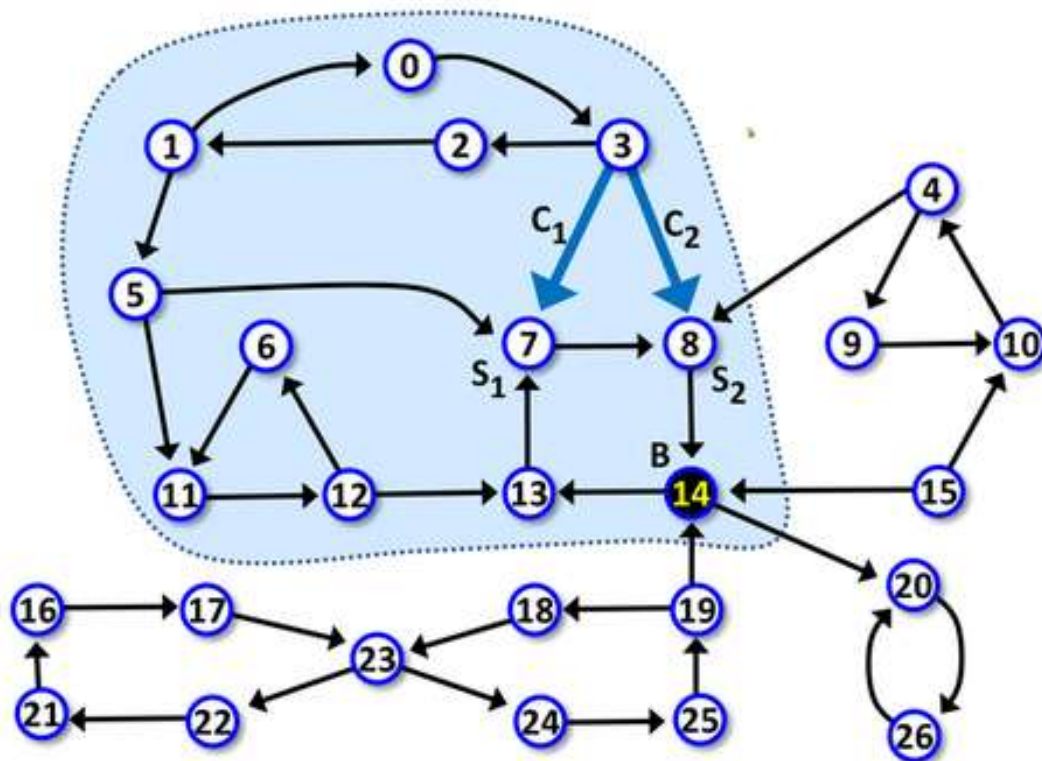


Image 1. The scheme depicts canal layouts in Lake district. Canals and their directions are depicted as arrows, lakes as numbered circles. When the barge starts its journey in one of the reversed canals C_1 , C_2 , it will be able to access all 12 lakes in the

highlighted area. This choice is optimal. Note that both possible start lakes S_1 and S_2 can be reached from lake B containing barge dock, and that also lake B can be reached from both possible start lakes S_1 , S_2 . The scheme depicts Example 1 below.

The task

You are given a scheme of the lakes and canals in Lake district and a specification of the lake with the barge dock. Calculate the maximum number of lakes which can be visited during one journey along the waterways, provided that the decision on choosing a canal to be temporarily reversed was done optimally.

Input

The first input line contains three positive integers LA , CA , DO , separated by spaces, where LA represents the number of lakes, CA represents the number of canals, and DO is the label of the lake with the barge dock. The lakes are labeled $0, 1, \dots, LA-1$. Next, there are CA text lines, each specifies one canal. The line contains two integers a, b separated by space, a is the label of the start lake of the canal, b is the label of the end lake of the canal. The canals are listed in no particular order.

It holds, $4 < LA \leq 4 \times 10^5$, $4 < CA \leq 10^6$.

Output

The output consists of a single text line containing one integer equal to the maximum number of lakes which can be visited during one journey through the given waterways.

It is not guaranteed that the canal direction reversal at the beginning of the barge journey increases the number of lakes the barge can visit, compared to a situation in which no canal would be reversed.

Example 1

Input

```
27 37 14
0 3
1 0
1 5
2 1
3 2
3 7
3 8
4 8
4 9
5 7
5 11
6 11
7 8
8 14
9 10
10 4
11 12
12 6
12 13
13 7
14 13
14 20
15 10
15 14
16 17
17 23
18 23
19 14
19 18
20 26
21 16
22 21
23 22
23 24
24 25
```

Example 2

Input

```
21 31 4
0 1
1 2
2 3
3 4
4 0
5 6
6 5
7 8
8 9
9 7
10 11
11 12
12 13
13 10
14 15
15 16
16 17
17 18
18 14
19 20
20 19
5 0
6 0
7 1
9 1
10 2
13 2
14 3
18 3
19 4
20 4
```

Output

Example 3

Input

```
11 15 1
0 1
1 4
4 0
2 3
3 4
5 4
5 6
6 10
10 5
9 10
9 8
8 7
7 2
3 8
9 4
```

Output

```
8
```

25 19
26 20

10

Output

12

Topographical note for curious

When observing two canals in Lake district from above, they may sometimes appear to be crossing each other. However, it is only a matter of perspective. In such situation the canals on the ground do not intersect, and they do not share any water. One canal is always positioned above the other one in the landscape, typically with the help of a special bridge. Without the bridges, the number of the canals in relation to the number of lakes (LA) would be limited by the value $6 \times LA - 12$, which is not the case for a typical canal layout in the country.

Public data

The public data set is intended for easier debugging and approximate program correctness checking. The public data set is stored also in the upload system and each time a student submits a solution it is run on the public dataset and the program output to stdout and stderr is available to him/her.

[Link to the public data set](#)